1 RELATIONS AND CONSTRAINTS

Every relation has an autogenerated `Id` INTEGER primary key. Foreign-key columns are INTEGER; `WeightFraction` is REAL; other columns are TEXT. Tables are STRICT. All columns are NOT NULL except those marked `?`; required text is nonempty, and identity text has no surrounding spaces. All FKs reference `Id`, with UPDATE/DELETE RESTRICT. **UK** denotes uniqueness; conditional UKs apply only to their stated rows.

Customer
`Id, CustomerNumber, GivenName, FamilyName, Email?`
UK: CustomerNumber. Names and Email are not unique.

Portfolio
`Id, CustomerId, PortfolioReference, PortfolioName, BaseCurrency`
FK: CustomerId → Customer. UK: (CustomerId, PortfolioReference).

Holding
`Id, PortfolioId, ETFId, WeightFraction, HoldingAsOfDate`
FKs: PortfolioId → Portfolio; ETFId → ETF. UK: (PortfolioId, ETFId, HoldingAsOfDate).

ETFProvider
`Id, LegalName, RegistrationNumber, RegistrationJurisdiction, WebsiteUrl?`
UK: (RegistrationJurisdiction, RegistrationNumber). LegalName is descriptive, not unique.

ETF
`Id, ETFProviderId, FundName, DomicileCountryCode, BaseCurrency, InceptionDate?`
FK: ETFProviderId → ETFProvider. Business identity is supplied by ETFIdentifier; FundName is not unique. Each saved ETF requires at least one identifier (transaction rule).

ETFIdentifier
`Id, ETFId, IdentifierScheme, IdentifierValue, ExchangeContext?`
FK: ETFId → ETF. IdentifierScheme ∈ {ISIN, Ticker}. Values are uppercase and nonempty. ISIN: UK (IdentifierScheme, IdentifierValue), ExchangeContext is NULL; value has 12 uppercase alphanumeric characters, two initial letters and a final digit. Ticker: UK (IdentifierScheme, IdentifierValue, ExchangeContext); exchange is required, uppercase and nonempty. ISIN checksum is not checked.

Common scalar rules. Fractions are in $[0, 1]$. Currency codes have three uppercase letters and domicile codes two. Dates are valid `YYYY-MM-DD`; source timestamps are UTC `YYYY-MM-DDTHH:MM:SS.ffffffZ`. Optional text is NULL when unknown and nonempty when present. Code formats do not verify external registry membership.

SourceDocument
`Id, ETFProviderId, Url, RetrievedAt, PublishedAt?, CapturedFileRef, DocumentTitle?`
FK: ETFProviderId → ETFProvider. UK: (Url, RetrievedAt). CapturedFileRef identifies a retained capture; the transaction validator checks file existence within the supplied project root.

AllocationDimension
`Id, DimensionName`
UK: DimensionName; DimensionName ∈ {Sector, Geography}.

ClassificationScheme
`Id, SchemeAuthority, SchemeName, Version?, DefinitionUrl?`
Conditional UK: (SchemeAuthority, SchemeName, Version) WHERE Version IS NOT NULL. Unknown editions retain NULL Version and no asserted complete business key.

AllocationCategory
`Id, AllocationDimensionId, ClassificationSchemeId?, Label, Code?, ContextKey`
FKs: AllocationDimensionId → AllocationDimension; ClassificationSchemeId → ClassificationScheme (optional). UK: (AllocationDimensionId, ContextKey, Label). ContextKey is generated from known authority/name/version or from captured URL/retrieval time/block locator/dimension. Unresolved contexts are local to the disclosure; matching labels/codes do not prove equivalence.

AllocationSnapshot
`Id, ETFId, AllocationDimensionId, SourceDocumentId, AllocationAsOfDate, BasisNote, CoverageStatus, SourceLocator`
FKs: ETFId → ETF; AllocationDimensionId → AllocationDimension; SourceDocumentId → SourceDocument. UK: (ETFId, AllocationDimensionId, AllocationAsOfDate, SourceDocumentId, SourceLocator). CoverageStatus ∈ {Complete, Partial, Unknown}. SourceLocator distinguishes dated allocation blocks/bases within a capture.

AllocationEntry
`Id, AllocationSnapshotId, AllocationCategoryId, WeightFraction`
FKs: AllocationSnapshotId → AllocationSnapshot; AllocationCategoryId → AllocationCategory. UK: (AllocationSnapshotId, AllocationCategoryId).

Cross-relation rules. SQL triggers require matching category/snapshot dimensions on entry writes and parent dimension edits. Before COMMIT, the transaction validator requires an entry per saved snapshot, validates used category contexts and checks Complete totals within 0.005 of 1 (inclusive, configurable). Partial/Unknown totals and sampled holding totals need not equal 1. Parent minimums, contexts, totals and file checks require the application validator; raw SQL COMMIT can bypass them.